Dipesh Dulal

nilkantha.dipesh@gmail.com
github.com/dipeshdulal
+977-9843435969
medium.com/@dipesh-dulal
Kathmandu, Nepal

Software engineer and engineering lead with 10 years of experience building web, mobile and cloud products — from Japanese e-commerce platforms and ISO 27001 compliance systems to developer tooling. I lead teams end-to-end: architecture, delivery, code review and developer experience. Currently focused on Go services on AWS, type-safe TypeScript frontends and AI-assisted engineering workflows.

Skills

- Languages – Go, TypeScript, JavaScript, Python, Dart, Java, PHP, SQL
- Backend – Gin, GORM, uber-go/fx, templ, Node.js, Next.js, Django, Laravel, Spring Boot, GraphQL
- Frontend/Mobile – React 19, TanStack, Tailwind, Radix/shadcn, TipTap, Flutter, React Native, Three.js
- Databases – PostgreSQL, MySQL, Atlas migrations
- Cloud/DevOps – AWS (Lambda, Fargate, Elastic Beanstalk, Cognito, S3, CloudFormation), GCP, Terraform, Pulumi, Docker, GitHub Actions, Cloudflare Workers, Tailscale
- Practices – Multi-tenant architecture, RBAC and permissions, audit logging, Playwright end-to-end testing, Sentry, LLM integration (OpenAI, Llama, MCP)

Education

- **Phoenix College of Management, Lincoln University** — Kathmandu, Nepal
  Master of Computer Science; GPA 3.94; Thesis: Serverless API generation with LLM agents — *2022 – 2024*
- **Advanced College of Engineering and Management, IOE, TU** — Lalitpur, Nepal
  Bachelor's in Computer Engineering; 80.32% — *2015 – 2019*

Experience

- **wesionaryTEAM** — Tokyo, Japan
  CTO — *Apr 2021 - Present*
  - Lead a 28-engineer technical org at a 34-person company (peak 60), delivering 10+ client products across commerce, real-time chat, compliance and access control.
  - Architect across 24 repositories: Go services on AWS (80+ Lambdas, 7 Elastic Beanstalk environments, Aurora, 70+ queues) and React 19 / TanStack monorepos.
  - Client portfolio: Cospa marketplace core (Rakuten, Tsuhan-Sozai, Shopify), Tsuhan-Sozai/Precocirico storefronts, Yomiuri real-time chat (Go, Flutter, React, AppSync/CDK), Carcamping (Stripe subscriptions, LINE login, maps), SOUDAN, an ISO 27001 platform and EES biometric access control.
  - Ship payment integrations and processor migrations across products: ZEUS and DGFT for Tsuhan-Sozai, Stripe subscriptions for Carcamping, plus refunds, cancellations and fallback paths.
  - Design permission, approval and audit systems for regulated workflows: granular permissions, staged and proxy approvals, re-approval triggers and immutable audit trails.
  - Own security and incident response across the org: policy and severity model, RCA process (120 documented incidents), trusted proxies, CVE and supply-chain response, audit-log conformance.
  - Run R&D and prototyping: evaluate new stacks and build POCs for LLM/AI features, MCP tooling and agent-driven code review, turning findings into production architecture.
  - Run production for the largest account (8,900+ accounts, 8M requests/month, 115 req/s peak, 108ms p95); cut dev AWS spend 61% ($345/month) and set a $2.4k to $1.2k/month production target.
  - Hire and mentor: interviewed 100+ candidates and hired around 10; review 2,200+ PRs since 2022 and mentor engineers across teams on architecture and design.

- **wesionaryTEAM** Tokyo, Japan
  Senior Software Engineer *June 2020 - Apr 2021*
  - Built and shipped full-stack features with Go, React/Next.js, Flutter and React Native across client projects spanning commerce, health and media.
  - Delivered marketplace and master-data integrations across Rakuten, Tsuhan-Sozai and Shopify, covering orders, multi-warehouse inventory, carrier tariffs and Google Sheets sync.
  - Containerized services and built deployment pipelines for AWS, GCP, Play Store and App Store; researched and documented new technologies through POCs.
- **Phoenix Solutions** Maitidevi, Nepal
  Full-Stack Developer *Jan 2019 - May 2020*
  - Full Stack Software Development using Spring Boot, Laravel, React, React Native Technologies
  - Maintaining in built systems in Angular, Spring Boot and added some features
  - Building form builder using Angular JIT and self made template engine
  - Research on ERP Systems and no code/low code approach to system design and development using Idempiere
  - Mobile application design and development.
  - Deployment of developed systems to in-premise server.
  - Containerize applications, Wrote scripts for deployment and research on continuous integration and continuous deployment using Jenkins
- **Budhanilkantha Sanchar Griha** Budhanilkantha, Nepal
  Web Developer/Administrator *Nov 2016 - Dec 2018*
  - Create website bsgnews.com from planning, requirement definition, development, deployment to support.
  - Prepare Designs for social media posts, website and flex board's

Selected Technical Work

- **Order Allocation Engine — Cospa Creation**: Rules-based engine that decides which warehouse, vendor and carrier fulfils each order: FIFO across good and expected stock, row-locked transactions for concurrent orders, deallocation and reallocation, plus direct-sales and work-request paths, run as asynchronous Lambda workers — Go, PostgreSQL, AWS
- **Device Fleet Management — Entry Exit Management System**: Command queue for biometric terminals: batched fan-out of face and palm templates, OTA-safe S3 archiving (Parquet for Athena, JSONL for restore), and a verification model that re-reads templates from the device because acknowledgements lie. Built a push simulator to reproduce protocol races — Go, PostgreSQL, Next.js 15
- **ISO 27001 Compliance Platform**: Multi-tenant ISMS platform for a Japanese certification body: granular permission model, staged and proxy approvals, re-approval rules, immutable audit trail, certification registers and EN/JA report catalogue — 176k lines of Go, 133 migrations, 96 screens, 283 test files — Go, React 19, TanStack, PostgreSQL, AWS
- **Multi-Marketplace Commerce Core**: Rebuilt and migrated a legacy PHP core system to Go/AWS with a 15-minute cutover and full backward compatibility, then shipped on zero-downtime rolling deployments: orders, inventory and master-data sync across Rakuten, Tsuhan-Sozai and Shopify, a sheet catalogue, diff-file compares and multi-warehouse tariffs — 87 Lambdas, 7 Elastic Beanstalk environments — Go, templ, TypeScript, AWS

Earlier projects: MetaPylon Studio (Three.js/Web3, ExpMap projection for 3D sticker placement), Health Journey (Go/React PHR), GLIVE (WebRTC golf streaming), KOL (influencer marketing), SCM for Otech Traders (iDempiere ERP), Appointment Management (React Native/Laravel), Municipal Planning (Spring Boot/Angular, Nepal Government).

Open Source & Hardware

- Gothalo – self-hosted mobile remote for coding agents: a Go bridge and Flutter app with live terminal, push notifications and Tailscale-only networking; 400+ commits.
- clean-gin – Go clean-architecture template (Gin, uber-go/fx, GORM) used as a reference by other teams; 500+ GitHub stars.
- HWCX USB Display Driver – reverse-engineered a proprietary USB HID protocol and wrote a Linux display driver.
- Darkflash DN360D LED Driver – reverse-engineered the AIO cooler's LED controller protocol and built a Linux driver.